



Learn how to tune your hard disk I/O using `read_ahead`, elevators, queuing theory and the utilisation law.

Performance Tuning and Monitoring

Last month we learned how to check the capabilities of our existing hardware, and how to ascertain the current load on the system using tools like *iostat*, *dmidecode*, *sar*, *mrtg*, *gnuplot*, etc. Well, it's the right time to tune your system to your needs, after monitoring the data now available to you.

First, tune your HDD I/O.

Tuning disk I/O using elevators

Hard disks are electro-mechanical devices with two kinds of components—internal and external. While the internal component is essentially the sealed chamber called the hard drive assembly (HDA), the external components are located on a printed circuit board (PCB), often referred as the logic board. Refer to Figures 1 and 2.

Today's hard disk drives are made of platters coated with magnetic material. An electromechanically actuated armature contains the 'read/write' heads that move between these platters. The armature is normally composed of multiple arms. All the arms are bound together to work as a single unit. So when the armature moves to a specific track on the platter, all the read/write heads are placed on the same track on each platter. This is known as a 'cylinder'.

Remember that whenever there is a moving part involved in I/O, there will be latency. As our

HDD also has moving parts—the platters and the armature—we cannot expect a 'wire speed' I/O. So whenever the disk controller requests a read/write from a particular sector, the armature has to move from its current position to the new requested position. This delay is known as the seek time. Add the time the motor takes to make that particular sector appear under the read/write head to the seek time—this time is called the rotational delay, or 'rotational latency'.

The good news is that all modern HDDs (more than 90 per cent of all known brands) use a concept called ZCAV (Zonal Constant Angular Velocity). This takes advantage of the fact that more linear space is available on the outer tracks of the disk platter rather than on the inside tracks. Now since the disk spins at a constant speed, which is also known as CAV (Constant Angular Velocity), the read/write I/O speed will be greater at the outer tracks as compared to the inner tracks. You can use a program called *Bonnie++* [www.coker.com.au/bonnie++] to check your hard disk.

So the general rule is that you should always create the locations that need frequent I/O—for example, `/home` and `swap`—on the outer tracks. Since generally all HDDs allot partitions from the outside, the easiest way to achieve this is to create these partitions first when partitioning your hard disk.